Supplementary information

Characterisation and Quantification of Data Centre Flexibility for Power System Support

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This document supplies additional input records, annual and event procedures, numerical checks and investment assumptions. The main paper retains its IT, UPS and cooling formulation and existing descriptive material. Tables of inputs and outputs accompany the source package as machine-readable CSV files. All equations describe the model used for the results with the corrected hourly tranche mapping.

S1 Workload and electricity-price inputs

S1.1 Adopted workload arrays and their interpretation

Published studies describe differing utilisation levels, workload classes and temporal flexibility across facilities [1, 2, 4, 6]. They motivate separating immediate work from work that may be rescheduled, but do not identify a universal flexible/inflexible ratio. The inputs used here are stylised profiles selected with reference to that literature. They are not claimed to be a measured site trace, a statistically representative sample of facilities or a deterministic reconstruction of a particular published trace.

The numerical inputs are `data/load_profiles.csv` and `data/shiftability_profile.csv`, whose active slots 1–96 are reproduced jointly in Table S1. The load file also contains legacy rows 97–108; annual timeline construction does not use those rows. The former file specifies flexible and inflexible arrival utilisation at quarter-hour resolution. The main paper's workload table retains hourly means of these inputs: for example, flexible utilisation at 00:00, 00:15, 00:30 and 00:45 is 0.40, 0.38, 0.36 and 0.34, so its hourly mean is 0.37 rather than the hourly starting value of 0.40. The four individual values are used by the model. The contextual literature-range columns do not replace these exact inputs.

The tranche-share file is constructed from the 24 hourly distributions in the main paper's tranche table. Each hourly row is repeated four consecutive times: the 00:00 distribution applies at 00:00, 00:15, 00:30 and 00:45, followed by four copies of the 01:00 distribution, and so on. This gives 96 quarter-hour rows, each summing to one. The whole 24-row sequence is not repeated four times across the day. Table S1 and the supplied CSV reproduce the intended expansion exactly.

For an arrival in interval t, class k receives work

$$a_{t,k} = u_t^{\text{flex,arr}} \alpha_{t,k} \Delta t, \quad \ell_{t,k} = t + D_k \Delta t, \quad (D_1, D_2, D_3, D_4) = (2, 4, 8, 12). \tag{S1}$$

The final execution interval starts at $\ell_{t,k}$ and closes at $\ell_{t,k} + \Delta t$. For example, a three-hour class arriving at 12:00 may execute in the interval starting at 15:00 and ending at 15:15. Thus the class names denote shifts between interval starts, not a guarantee of completion within exactly three continuous hours of an interval's opening. Cohort creation omits quantities no larger than the 10^{-7} CPU-hour numerical tolerance.

For workload multiplier m, let $v_t = u_t^{\text{inflex}} + u_t^{\text{flex,arr}}$ in the original input. The sensitivity uses

$$u_t^{\text{flex,arr}}(m) = \min\{m u_t^{\text{flex,arr}}(1), v_t\}, \qquad u_t^{\text{inflex}}(m) = v_t - u_t^{\text{flex,arr}}(m). \tag{S2}$$

Tranche fractions remain fixed. The clipping rule preserves total arriving CPU-hours and avoids creating more flexible work than the total workload. A multiplier is therefore not always an exactly proportional increase in the realised flexible share.

Table S1: Actual quarter-hour workload inputs. Utilisations are fractions of rated CPU capacity; tranche shares are fractions of flexible work.

Local start	Inflexible	Flexible	α_1	α_2	α_3	α_4
00:00	0.28	0.40	0.25	0.25	0.20	0.30
00:15	0.27	0.38	0.25	0.25	0.20	0.30
00:30	0.26	0.36	0.25	0.25	0.20	0.30
00:45	0.25	0.34	0.25	0.25	0.20	0.30
01:00	0.25	0.31	0.25	0.25	0.15	0.35
01:15	0.23	0.32	0.25	0.25	0.15	0.35
01:30	0.21	0.32	0.25	0.25	0.15	0.35
01:45	0.19	0.32	0.25	0.25	0.15	0.35
02:00	0.17	0.33	0.35	0.17	0.18	0.30
02:15	0.16	0.30	0.35	0.17	0.18	0.30
02:30	0.16	0.28	0.35	0.17	0.18	0.30
02:45	0.16	0.26	0.35	0.17	0.18	0.30
03:00	0.16	0.23	0.25	0.15	0.20	0.40
03:15	0.14	0.24	0.25	0.15	0.20	0.40
03:30	0.12	0.25	0.25	0.15	0.20	0.40
03:45	0.10	0.26	0.25	0.15	0.20	0.40
04:00	0.08	0.27	0.25	0.20	0.15	0.40
04:15	0.07	0.27	0.25	0.20	0.15	0.40
04:30	0.06	0.27	0.25	0.20	0.15	0.40
04:45	0.06	0.27	0.25	0.20	0.15	0.40
05:00	0.06	0.27	0.25	0.15	0.15	0.45
05:15	0.07	0.24	0.25	0.15	0.15	0.45
05:30	0.09	0.22	0.25	0.15	0.15	0.45
05:45	0.10	0.20	0.25	0.15	0.15	0.45
06:00	0.12	0.18	0.23	0.28	0.15	0.34
06:15	0.14	0.19	0.23	0.28	0.15	0.34
06:30	0.16	0.21	0.23	0.28	0.15	0.34
06:45	0.18	0.22	0.23	0.28	0.15	0.34
07:00	0.20	0.24	0.20	0.20	0.15	0.45
07:15	0.21	0.24	0.20	0.20	0.15	0.45
07:30	0.22	0.24	0.20	0.20	0.15	0.45
07:45	0.23	0.24	0.20	0.20	0.15	0.45
08:00	0.24	0.24	0.25	0.15	0.13	0.47
08:15	0.26	0.25	0.25	0.15	0.13	0.47
08:30	0.29	0.26	0.25	0.15	0.13	0.47
08:45	0.31	0.27	0.25	0.15	0.13	0.47
09:00	0.34	0.28	0.25	0.17	0.16	0.42
09:15	0.35	0.25	0.25	0.17	0.16	0.42
09:30	0.36	0.23	0.25	0.17	0.16	0.42
09:45	0.37	0.21	0.25	0.17	0.16	0.42
10:00	0.37	0.19	0.35	0.22	0.13	0.30
10:15	0.38	0.18	0.35	0.22	0.13	0.30

Local start	Inflexible	Flexible	α_1	α_2	α_3	α_4
10:30	0.39	0.17	0.35	0.22	0.13	0.30
10:45	0.40	0.16	0.35	0.22	0.13	0.30
11:00	0.42	0.16	0.32	0.20	0.15	0.33
11:15	0.41	0.17	0.32	0.20	0.15	0.33
11:30	0.40	0.18	0.32	0.20	0.15	0.33
11:45	0.39	0.19	0.32	0.20	0.15	0.33
12:00	0.40	0.20	0.45	0.25	0.15	0.15
12:15	0.39	0.21	0.45	0.25	0.15	0.15
12:30	0.39	0.22	0.45	0.25	0.15	0.15
12:45	0.38	0.23	0.45	0.25	0.15	0.15
13:00	0.36	0.24	0.50	0.20	0.15	0.15
13:15	0.36	0.24	0.50	0.20	0.15	0.15
13:30	0.35	0.25	0.50	0.20	0.15	0.15
13:45	0.35	0.26	0.50	0.20	0.15	0.15
14:00	0.35	0.27	0.40	0.20	0.25	0.15
14:15	0.34	0.27	0.40	0.20	0.25	0.15
14:30	0.33	0.27	0.40	0.20	0.25	0.15
14:45	0.32	0.27	0.40	0.20	0.25	0.15
15:00	0.33	0.27	0.45	0.25	0.15	0.15
15:15	0.34	0.25	0.45	0.25	0.15	0.15
15:30	0.35	0.23	0.45	0.25	0.15	0.15
15:45	0.37	0.21	0.45	0.25	0.15	0.15
16:00	0.40	0.20	0.50	0.15	0.20	0.15
16:15	0.37	0.25	0.50	0.15	0.20	0.15
16:30	0.33	0.30	0.50	0.15	0.20	0.15
16:45	0.30	0.35	0.50	0.15	0.20	0.15
17:00	0.27	0.40	0.50	0.20	0.15	0.15
17:15	0.27	0.41	0.50	0.20	0.15	0.15
17:30	0.27	0.42	0.50	0.20	0.15	0.15
17:45	0.26	0.43	0.50	0.20	0.15	0.15
18:00	0.26	0.45	0.10	0.20	0.30	0.40
18:15	0.26	0.45	0.10	0.20	0.30	0.40
18:30	0.26	0.45	0.10	0.20	0.30	0.40
18:45	0.26	0.45	0.10	0.20	0.30	0.40
19:00	0.26	0.45	0.15	0.20	0.25	0.40
19:15	0.26	0.45	0.15	0.20	0.25	0.40
19:30	0.25	0.46	0.15	0.20	0.25	0.40
19:45	0.25	0.46	0.15	0.20	0.25	0.40
20:00	0.25	0.47	0.18	0.12	0.25	0.45
20:15	0.26	0.46	0.18	0.12	0.25	0.45
20:30	0.27	0.45	0.18	0.12	0.25	0.45
20:45	0.28	0.44	0.18	0.12	0.25	0.45
21:00	0.29	0.41	0.22	0.18	0.25	0.35
21:15	0.29	0.41	0.22	0.18	0.25	0.35
21:30	0.29	0.41	0.22	0.18	0.25	0.35
21:45	0.30	0.41	0.22	0.18	0.25	0.35
22:00	0.30	0.40	0.20	0.16	0.24	0.40
22:15	0.27	0.40	0.20	0.16	0.24	0.40
22:30	0.25	0.41	0.20	0.16	0.24	0.40
22:45	0.22	0.41	0.20	0.16	0.24	0.40

Local start	Inflexible	Flexible	α_1	α_2	α_3	α_4
23:00	0.21	0.42	0.15	0.20	0.20	0.45
23:15	0.21	0.42	0.15	0.20	0.20	0.45
23:30	0.21	0.42	0.15	0.20	0.20	0.45
23:45	0.21	0.42	0.15	0.20	0.20	0.45

S1.2 Prices, calendar and settlement

The source is LCCC's IMRP actuals dataset [3], whose reference product is the Great Britain day-ahead hourly price. Source fields identify local date, chronological hourly settlement period and IMRP amount in GBP/MWh. The 2025 series covers 8,760 physical hours. Hourly prices are replicated to four consecutive quarter-hours without smoothing, giving 35,040 committed intervals. The timeline uses UTC for physical ordering and Europe/London for local-date and workload-slot labels. Spring and autumn clock changes produce 92 and 100 quarter-hours in those local days; workload values for skipped/repeated local-clock periods are correspondingly skipped/repeated. No physical UTC interval is missing or duplicated.

Negative prices remain signed in both optimisation and reported costs. There are 672 negative-price quarter-hours in the retained annual data. IMRP is a reference-price signal, not a full retail tariff. The first three hourly prices on 1 January 2026 are GBP 46.49, 40.19 and 28.99/MWh; they supply the twelve final look-ahead intervals. The exact expanded central price timeline is included as `data/price_timeline.csv`. Annual cost uses only committed intervals in calendar 2025.

S2 Annual evaluation and state transfer

Each annual case follows this procedure:

1. Build the continuous UTC price timeline and assign Europe/London date and profile-slot labels. Load the quarter-hour workload and tranche arrays.

2. Initialise UPS/TES energy and the four thermal states once, at the beginning of the year; the initial unfinished-work ledger is empty.

3. For local day d, select its 92, 96 or 100 core intervals and append twelve further intervals. Load the carried state and unfinished cohorts, and generate new arrivals throughout the horizon.

4. Solve the reference or coordinated model, and record feasibility, objective bounds and accounting checks. Both core and look-ahead contain the full facility demand.

5. Commit only core controls and processing. Accumulate signed-price settlement for those intervals, save their ending physical state and compute unfinished work from committed processing only.

6. Pass the ending state and unfinished cohorts to the next local day. Reoptimise all future controls. Repeat through 31 December.

In this supplement, $\mathcal{C}_d = T_d$ and $\mathcal{H}_d = T_d^{\text{ext}}$ in the main-paper notation. A cohort c identifies one arrival/tranche pair (t, k); r_c and ℓ_c are its arrival and final eligible interval start, $A_{c,d}$ its available work, and $R_{c,H}$ its residual at boundary H. Superscript op denotes Scenario 2, and ref denotes Scenario 1. Powers and stored energies use the main-paper units.

If $\boldsymbol{x}$ contains the two energy states and four temperatures, physical transfer is $\boldsymbol{x}_{d+1}^{\text{start}} = \boldsymbol{x}_d^{\text{core,end}}$. There is no daily storage equality restoring the initial state. Storage and workload

transfer are separate: an energy state alone cannot reconstruct which jobs remain eligible for execution.

For annual committed set $\mathcal{Y}$, reported settlement is

$$C = \sum_{t \in \mathcal{Y}} \pi_t P_t^{\text{grid}} \Delta t / 1000. \tag{S3}$$

This counts every 2025 interval once. The final continuation contains ordinary January demand as well as outstanding December work, so its entire cost is not added to 2025 expenditure. The central ledger contains 0.787115 CPU-hours at midnight after 31 December, and the retained continuation serves that work within its permitted execution windows. The final continuation's feasibility does not imply that its full dispatch is committed as an additional annual day.

The implementation records each day's controls, physical states, cohort residuals, solver status and audits. Central operation also records actual processing by cohort and interval, permitting event initialisation from the exact unfinished-work ledger. Recorded daily objectives include look-ahead decisions, whereas reported annual costs include committed decisions only. A daily MILP gap is therefore not an optimality certificate for a single full-year optimisation problem.

S3 Representative-day and flexibility-event procedures

S3.1 Day selection

Daily features from central coordinated operation are: mean price, price standard deviation, minimum and maximum price, negative-price share, mean and peak grid import, mean total CPU utilisation, daily settlement cost, and opening UPS, TES and cold-aisle states. Each feature is centred on its mean across all annual daily observations and scaled by its population standard deviation; a zero standard deviation is replaced by one. For candidate day d, the score is

$$s_d = \left[\frac{1}{12} \sum_{j=1}^{12} \left(\frac{v_{dj} - \bar{v}_j}{\sigma_j} \right)^2 \right]^{1/2}. \tag{S4}$$

Candidates have 96 intervals and local date before 29 December 2025. The smallest score is selected, with date used to break a tie. This gives 22 September, score 0.23481769. The complete candidate table accompanies the package. The rule identifies a central multivariate illustration; it does not weight days by event availability, or guarantee a typical duration envelope.

S3.2 Opening state and recovery

For each hourly t_0, load the undisturbed central trajectory and exact remaining work at that boundary. Pre-event execution is fixed by the stored history; the event solve starts at t_0. New arrivals from that point are generated with their original class rules. Any existing cohort retains only its remaining work and remaining eligible intervals.

For duration τ, the trial ends at $H = t_0 + \tau + 3$ h. Impose a grid deviation of ΔP within ± 0.1 kW during the event, with no grid-target constraint during the recovery period. At H, require UPS/TES energy no smaller than the undisturbed states, each temperature within 0.05°C of its undisturbed value, and each cohort residual no larger than the undisturbed residual plus 10^{-7} CPU-hours. Workload eligibility, storage modes and all physical limits hold throughout. Surplus stored energy is permitted at recovery; exact equality of storage states is not required.

S3.3 Search, certification and dispatch interpretation

There are 24 start times, nine negative requests (−100 to −500 kW in 50 kW steps) and three positive requests (25, 50, 75 kW). For each pair, the assessment cap in intervals is the smaller of 48 and the number of intervals remaining in the local day. The procedure first tests that cap. If unsuccessful, it searches logarithmically for a feasible duration and checks the following interval. Trials in this assessment have a 60 s solver limit; unresolved adjacent trials are retained as lower bounds.

A trial with an audited feasible incumbent is evidence that its duration can be delivered under the model. A trial without an incumbent can be either proven infeasible or unresolved; these outcomes are retained separately. During the search, an unsuccessful trial can narrow the region explored even when it is unresolved. Consequently the results should be interpreted as the feasible durations found by this procedure, rather than as an exhaustive exploration of every duration.

The output flag `boundary_verified` covers both a feasible assessment cap and a proven-infeasible adjacent candidate. It is not, by itself, a proof that all longer durations are infeasible. Since extending an event changes the recovery boundary and undisturbed terminal reference, monotonic feasibility with duration has not been established for this formulation. The main paper therefore reports certified feasible durations. It retains plus signs for the eight unresolved adjacent cases and explains the cap convention separately. Unmarked cells do not acquire a stronger global-maximality interpretation merely because the next tested interval was infeasible. Blank cells mean that this search certified no positive duration.

The accepted dispatch minimises signed event-plus-recovery electricity cost subject to the request. Where a solve terminates with a nonzero objective gap, it is an accepted incumbent for that objective, not a proven minimum-cost allocation. Even a proven optimum need not be unique. Thus component panels illustrate feasible coordinated mechanisms and exact electrical accounting; they are neither a unique attribution nor an independent valuation of the assets. The dashed target ends at the certified event duration. Grey regions contain no plotted recovery dispatch. A plus sign beside a duration denotes an unresolved adjacent trial.

S4 Extended annual and component results

All sensitivity points in Table S2 are completed annual chains. They supersede earlier representative-week intermediate comparisons. The common reference has the same total workload arrivals as every flexible-share case; storage is disabled in that reference. Percentage-point changes are relative to the central coordinated saving, while saving percentages use the common reference cost as denominator. More decimals are retained in the accompanying CSV files for calculation.

The annual endpoints and event grid are supplied as `data/annual_endpoints.csv` and `data/flexibility_results.csv`. The representative-day component schedule remains in the main paper.

The annual component-cost differences are accounting terms within the same coordinated solution. In particular, electricity used to charge TES is positive in the charging-chiller row, while avoided direct cooling appears elsewhere. The UPS row is charging minus discharging evaluated at interval prices, not a storage-revenue settlement. The accounting identity remains valid under negative prices. Summed displayed values can differ from displayed totals by rounding; unrounded terms reconcile in the accompanying data.

Table S2: Full-year cost and resource sensitivity. Change is percentage points from the central coordinated saving; baseline denotes reference operation.

Case	Cost (GBP)	Saving (%)	Change (pp)	Energy (GWh)
Baseline	550,504	0.000	-5.382	6.718
Flexible workload 0.5×	529,934	3.737	-1.645	6.777
Flexible workload 0.75×	525,157	4.604	-0.778	6.781
Flexible workload 1.0×	520,877	5.382	+0.000	6.789
Flexible workload 1.25×	516,997	6.087	+0.705	6.797
Flexible workload 1.5×	513,989	6.633	+1.251	6.806
UPS capacity 0.5×	523,544	4.897	-0.484	6.766
UPS capacity 0.75×	522,187	5.144	-0.238	6.778
UPS capacity 1.25×	519,602	5.613	+0.232	6.799
UPS capacity 1.5×	518,417	5.829	+0.447	6.810
TES capacity 0.5×	522,268	5.129	-0.253	6.779
TES capacity 0.75×	521,496	5.269	-0.113	6.784
TES capacity 1.25×	520,502	5.450	+0.068	6.792
TES capacity 1.5×	520,229	5.499	+0.118	6.794

S5 Numerical checks and solution quality

The retained manuscript run record identifies Python 3.12.0, Pyomo 6.10.1 and HiGHS 1.15.1, on Windows 11 with an eight-core AMD Ryzen 7 9800X3D. Annual scenarios request a relative MILP gap of 0.1%, with a standard daily time limit of 300 s and a normal acceptance threshold of 1%. Feasible exceptions in sensitivity cases are disclosed below rather than treating every solve as proven optimal. The main paper's 5.382% is computed from the retained annual settlements, not by summing daily objective bounds.

All fourteen annual summaries contain 365 local dates and 35,040 committed intervals. Their stored checks cover timestamp continuity, physical state handoff, workload conservation, storage mode exclusivity and signed-price settlement reconciliation. The stated audit tolerances are 10^{-5} in the native state units, 10^{-3} kW for power flows and 10^{-7} CPU-hours for work. Central operation also stores a cohort execution trace. The three central time-limited daily solutions, on 25 May, 6 August and 4 October, have reported relative gaps of approximately 0.868%, 0.228% and 0.612%, respectively, within the normal threshold.

Table S3: Corrected annual-chain solver records. Non-optimal denotes the stored termination classification; each listed chain covers all 365 days.

Case	Non-optimal horizons	Maximum recorded gap (%)
Baseline	0	0.000
Flexible workload 0.5×	1	1.301
Flexible workload 0.75×	1	0.509
Flexible workload 1.0×	3	0.868
Flexible workload 1.25×	4	0.686
Flexible workload 1.5×	1	0.496
UPS capacity 0.5×	2	1.111
UPS capacity 0.75×	2	0.684
UPS capacity 1.25×	2	0.554
UPS capacity 1.5×	2	0.475
TES capacity 0.5×	2	1.009
TES capacity 0.75×	2	0.894
TES capacity 1.25×	3	0.572
TES capacity 1.5×	3	0.903

Table S3 records the maximum reported gap for each retained annual chain. Three 25 May sensitivity horizons exceed the normal 1% threshold: the 0.5 workload, 0.5 UPS and 0.5 TES cases have gaps of 1.301%, 1.111% and 1.009%, respectively. Their absolute daily objective-bound widths are GBP 0.103, 0.329 and 0.359. These feasible time-limited incumbents remain in the annual settlements and are disclosed as exceptions. Near-zero negative objectives can produce relative gaps above the threshold despite small absolute bounds. The accompanying `data/corrected_solver_horizons.csv` records every non-optimal termination in the corrected annual chains.

The daily bounds apply to the corresponding core-plus-look-ahead optimisation, not to a globally optimised annual problem. They do not bound the downstream effect of different committed decisions or certify an annual saving to the displayed rounding precision. Historical longer-limit diagnostic solves used the earlier tranche inputs and are not used to certify these corrected results. In particular, the event study's 1% criterion must not be interpreted as a claim that all annual horizons meet that threshold.

The corrected central year-end checkpoint records 0.787115 CPU-hours outstanding at midnight and zero unserved work after the planned continuation. Its complete final look-ahead contains 2,531.359 kWh and GBP 98.275, including ordinary January demand. These quantities come from the current checkpoint supplied in `evidence/corrected_year_end_checkpoint.json`. Earlier terminal-value tests used different tranche inputs; their report is retained as historical context for the terminal convention, not as a terminal-sensitivity validation of the corrected central or enlarged-capacity cases.

For the event study, all 288 cells pass the recorded validation checks: 280 carry the boundary/cap flag and eight an unresolved-next-step flag. The 148 positive-duration dispatches meet the event solver-gap criterion of 1% and the physical audits; the other 140 cells have no certified positive duration. Maximum target error is 0.1 kW, terminal-temperature error 0.05°C, excess cohort residual 10^{-7} CPU-hours and component-sum residual approximately 4.82×10^{-11} kW. An independent accounting check reconciles all positive-duration component traces against metered deviations. These checks establish consistency with the formulated constraints. They do not eliminate the search qualifications in S3, model approximation or uncertainty in the adopted facility parameters.

S6 UPS break-even investment calculations

Let the additional annual electricity-cost saving for energy capacity E be $\Delta S(E) = C^{\mathrm{op}}(600) - C^{\mathrm{op}}(E)$. The retained results give GBP 1,274.7485/year for $E = 750$ kWh and GBP 2,459.8888/year for $E = 900$ kWh. Relative to the central capacity, the additional nameplate energies are 150 and 300 kWh, but the additional energy between operational bounds is 75 and 150 kWh because the 50% reserve fraction remains fixed.

For constant real annual benefits, additional investment I and additional annual non-electricity costs O, the investment breaks even when

$$\Delta S - O - I\,\mathrm{CRF}(r,n) \geq 0, \qquad \mathrm{CRF}(r,n) = \frac{r(1+r)^n}{(1+r)^n - 1}. \tag{S5}$$

At $r = 0$, CRF is $1/n$. The capital recovery factor converts initial expenditure into an equivalent annual amount [5]. Setting $O = 0$ gives $I_{\mathrm{BE}} = \Delta S/\mathrm{CRF}$. Dividing by added *nameplate* kWh gives a threshold directly comparable with a cost expressed on that same basis. The ten-year, 8% case has CRF 0.14902949. Table S4 shows additional illustrative rate/life combinations; these are assumptions for interpreting the result, not empirical estimates of project finance or battery life.

No battery market price is assumed, so the calculation establishes a maximum supported expenditure rather than a positive investment recommendation. Comparing a vendor quotation

Table S4: Illustrative supported investment per added nameplate kWh, in GBP/kWh, before additional non-electricity costs.

Life (years)	Real rate (%)	CRF	750 kWh case	900 kWh case
5	4	0.22463	37.83	36.50
5	8	0.25046	33.93	32.74
5	12	0.27741	30.63	29.56
10	4	0.12329	68.93	66.51
10	8	0.14903	57.02	55.02
10	12	0.17698	48.02	46.33
15	4	0.08994	94.49	91.17
15	8	0.11683	72.74	70.18
15	12	0.14682	57.88	55.85

would require its currency, price year, chemistry, installed scope, integration work, reserve policy and usable/nameplate capacity convention. Existing power electronics are held fixed in the dispatch sensitivity; an upgrade requiring additional converter capacity cannot be evaluated by changing only a battery price per kWh.

Losses already enter the dispatch costs through efficiency and must not be charged twice. Degradation, additional O&M, replacements and financing/tax detail are omitted. Introducing additional costs reduces supported investment under the same annual-benefit assumption. A more complete lifetime treatment could also change the benefit itself. In particular, initial UPS energy is proportional to capacity and terminal energy has no credit. The retained annual cases finish with 300, 375 and 450 kWh for 600, 750 and 900 kWh capacity, respectively. The resulting single-year differences are not independently established steady-state annual benefits. The earlier central terminal check in S5 does not resolve this for all enlarged-capacity cases.

Accordingly, repeating the 2025 incremental saving over n years is an explicit conditional assumption. The thresholds are conditional break-even investment estimates before omitted costs, not forecasts of a site's investment return. They exclude the possible value of greater backup provision and any revenues from grid services. Neither the event heatmap nor the annual component accounting supplies those revenues. If credible installed incremental costs exceed the stated threshold, the assumed energy-arbitrage saving alone is insufficient; if they are lower, the investment is supported only within the stated operational and financial assumptions.

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